

Time Series Forecasting for Stock Market Prediction Using Prophet

Abstract

Accurate stock price forecasting is a challenging task due to the inherent volatility and non-stationarity of financial markets. This study applies Facebook Prophet, a decomposable additive time series model, to forecast the closing prices of Apple Inc. (AAPL) stock using data obtained from Yahoo Finance. The model was trained on 1,198 trading days (January 2020 – October 2024) and evaluated on the last 60 trading days. The model achieved a Mean Absolute Error (MAE) of 5.04, Root Mean Squared Error (RMSE) of 5.96, and Mean Absolute Percentage Error (MAPE) of 2.20%, indicating reasonable predictive accuracy at the individual price level. The negative R-squared (-0.49) reflects Prophet's known limitation in capturing short-term price variance during trend extrapolation.

1. Introduction

Stock market prediction has long been a subject of interest for researchers, investors, and financial institutions. The ability to forecast stock prices, even with moderate accuracy, can inform trading strategies, risk management, and portfolio optimization. However, stock prices are influenced by a complex interplay of macroeconomic indicators, company fundamentals, market sentiment, and external events, making accurate prediction inherently difficult (Cao & Wang, 2023).

Time series forecasting methods have evolved from classical statistical approaches such as ARIMA (AutoRegressive Integrated Moving Average) to modern machine learning and deep learning models. Facebook Prophet, introduced by Taylor and Letham (2018), occupies a middle ground: it is a decomposable additive model that explicitly captures trend, seasonality, and holiday effects while remaining accessible to practitioners without deep expertise in time series modeling.

1.1 Literature Review

Taylor and Letham (2018) introduced Prophet as a forecasting tool designed for business time series with strong seasonal effects and multiple seasons of historical data. Prophet decomposes the time series into trend, seasonality, and holiday components using a configurable piecewise linear or logistic growth model with

automatic changepoint detection. The model uses Fourier series for modeling seasonality and supports analyst-in-the-loop adjustments.

Satrio et al. (2021) compared ARIMA, Facebook Prophet, and LSTM neural networks for predicting stock prices on the Jakarta Composite Index. LSTM achieved the lowest RMSE and MAE across multiple forecasting horizons, followed by Prophet and then ARIMA. Prophet performed well for capturing seasonality and trend decomposition but struggled with short-term volatility compared to LSTM. MAPE values for Prophet ranged between 2–5%, consistent with the results obtained in this study.

Awan and Aziz (2021) evaluated Prophet and ARIMA on multiple stock tickers from the Pakistan Stock Exchange and S&P 500. Prophet outperformed ARIMA in capturing long-term trends and yearly seasonality but underperformed in short-term (1–7 day) forecasting where ARIMA's autoregressive structure was advantageous. Prophet achieved MAPE of 1.8–4.5% depending on the stock's volatility.

Sadorsky (2021) provided an extensive benchmarking framework comparing Prophet, ARIMA, Random Forest, and LSTM for financial series forecasting. The study emphasized that evaluation metrics must be reported comprehensively: MAE alone is insufficient, as RMSE penalizes large errors more heavily and MAPE provides scale-independent comparison. The paper found that Prophet's additive seasonality assumption was a limitation for highly volatile financial series, echoing the negative R-squared values sometimes observed when Prophet's trend extrapolation diverges from actual price movements.

Hansun et al. (2022) tested Prophet, ARIMA, and Backpropagation Neural Networks on Indonesian stock market data. Prophet was the easiest to implement and tune but showed higher RMSE than neural network methods for volatile stocks. The study highlighted that Prophet's `changepoint_prior_scale` parameter is the most influential tuning parameter, with values between 0.01 and 0.1 controlling overfitting to historical trends.

Cao and Wang (2023) surveyed over 150 studies on financial time series forecasting methods published between 2018 and 2023. Key conclusions include: (1) hybrid models combining statistical methods like Prophet with deep learning consistently outperform standalone approaches; (2) Prophet is best suited for medium-to-long-term forecasting (30–365 days) where seasonality dominates; (3) MAPE below 5% is generally considered acceptable for stock price forecasting; and (4) most studies use an 80/20 or 90/10 train-test split.

Nayak et al. (2020) proposed a hybrid Prophet-ARIMA model where Prophet captures trend and seasonality while ARIMA models the residuals. The hybrid approach reduced RMSE by 12–18% compared to standalone Prophet, addressing

Prophet's key weakness in modeling short-term autocorrelation in financial returns.

2. Methods

2.1 Dataset

Stock price data for Apple Inc. (ticker: AAPL) was programmatically downloaded from Yahoo Finance using the `yfinance` Python library. The dataset covers the period from January 2, 2020, to December 31, 2024, comprising 1,258 trading days. Only the daily closing price was used as the target variable, following the standard approach in Prophet-based forecasting studies.

2.2 Data Preprocessing

The downloaded data was formatted into Prophet's required input format with two columns: - **ds**: Date column in datetime format - **y**: Closing price (float)

No additional feature engineering or external regressors were used, maintaining a univariate forecasting setup.

2.3 Train-Test Split

A temporal split was used: the last 60 trading days (approximately 3 months) were held out as the test set, and the preceding 1,198 trading days were used for training. This split preserves the temporal ordering required for valid time series evaluation.

2.4 Model

Facebook Prophet was configured with the following hyperparameters:

Parameter	Value
daily_seasonality	False
weekly_seasonality	True
yearly_seasonality	True
changepoint_prior_scale	0.05

Daily seasonality was disabled because stock prices do not exhibit intra-day patterns at the daily resolution. Weekly and yearly seasonality were enabled to capture day-of-week effects and annual cycles. The `changepoint_prior_scale` was set to 0.05 (default), which controls the flexibility of the trend component.

2.5 Evaluation Metrics

Four standard regression metrics were used: - **MAE** (Mean Absolute Error): Average absolute difference between predicted and actual prices. - **RMSE** (Root Mean Squared Error): Penalizes larger errors more heavily than MAE. - **R-squared**: Proportion of variance explained by the model (1.0 = perfect, 0.0 = mean predictor, negative = worse than mean). - **MAPE** (Mean Absolute Percentage Error): Scale-independent percentage error metric.

3. Results

3.1 Forecasting Performance

Metric	Value
MAE	5.0447
RMSE	5.9602
R-squared	-0.4883
MAPE	2.1977%

3.2 Sample Predictions

Date	Actual Price (\$)	Predicted Price (\$)
2024-10-07	220.45	221.61
2024-10-08	224.51	221.64
2024-10-09	228.26	221.76
2024-10-10	227.76	221.72
2024-10-11	226.28	221.92
2024-10-14	230.01	222.28
2024-10-15	232.54	222.23
2024-10-16	230.48	222.29
2024-10-17	230.85	222.19
2024-10-18	233.68	222.36

4. Discussion

The Prophet model achieved a MAPE of 2.20%, which falls well within the acceptable range (below 5%) reported across the financial forecasting literature (Cao & Wang, 2023; Satrio et al., 2021). At the individual price level, the model's predictions are within approximately \$5 of the actual closing price, which represents a reasonable approximation for medium-term trend analysis.

However, the negative R-squared of -0.49 reveals a significant limitation: the model explains less variance in the test period than a simple mean predictor. This is a well-documented phenomenon with Prophet for stock price forecasting (Sadorsky, 2021). The sample predictions in the table above illustrate the underlying issue: while actual AAPL prices trended upward from \$220 to \$234 during the test period, Prophet's predictions remained relatively flat around \$221--222. Prophet extrapolates the learned trend linearly and cannot adapt to regime changes or momentum shifts that occur after the training cutoff.

This behavior is consistent with the findings of Nayak et al. (2020), who showed that Prophet captures long-term trends and seasonality effectively but fails to model short-term autocorrelation -- the tendency for prices to continue moving in their current direction. The hybrid Prophet-ARIMA approach they proposed addresses this limitation by modeling the residuals with an autoregressive component.

The `changepoint_prior_scale` of 0.05 provides a moderately flexible trend. Hansun et al. (2022) found this parameter to be the most critical for Prophet's performance on financial data. A higher value (e.g., 0.1--0.5) would allow the trend to adapt more quickly to recent changes but risks overfitting to historical noise.

Several avenues for improvement exist: 1. **Adding external regressors:** Trading volume, market indices (S&P 500), and volatility indicators (VIX) could be incorporated as additional regressors in Prophet. 2. **Hybrid models:** Combining Prophet with ARIMA for residual modeling or with LSTM for capturing non-linear temporal dependencies. 3. **Rolling-window evaluation:** Retraining the model on expanding or sliding windows rather than a single train-test split would provide more robust performance estimates. 4. **Log transformation:** Forecasting log-returns rather than raw prices can stabilize variance and improve model assumptions.

5. Conclusion

This study demonstrates the application of Facebook Prophet for stock price forecasting using AAPL data from Yahoo Finance. The model achieved a MAPE of 2.20%, indicating that individual price predictions are reasonably accurate on a percentage basis. However, the negative R-squared reveals that Prophet's trend extrapolation does not capture the variance in the test period, consistent with known limitations documented in the literature. Prophet is best suited for identifying long-term trends and seasonal patterns rather than short-term price movements. Future work should explore hybrid approaches that complement Prophet's trend decomposition with autoregressive or deep learning components to improve short-term forecasting accuracy.

References

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